Use Cases

for

WalkingSG

Version 2.0 approved

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Group 3 – SCS2

September 20th

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Revision History

Name	Date	Reason For Changes	Version
Lin Ziqi	September 5th	Initial Use Case Model	1.0
Trinh Hoang Huy	September 5th	Final Touch for better look	2.0

Guidance for Use Case Template

Document each use case using the template shown in the Appendix. This section provides a description of each section in the use case template.

1. Use Case Identification

1.1. Use Case ID

Give each use case a unique numeric identifier, in hierarchical form: X.Y. Related use cases can be grouped in the hierarchy. Functional requirements can be traced back to a labeled use case.

1.2. Use Case Name

State a concise, results-oriented name for the use case. These reflect the tasks the user needs to be able to accomplish using the system. Include an action verb and a noun. Some examples:

- View part number information.
- Manually mark hypertext source and establish link to target.
- Place an order for a CD with the updated software version.

1.3. Use Case History

1.3.1 Created By

Supply the name of the person who initially documented this use case.

1.3.2 Date Created

Enter the date on which the use case was initially documented.

1.3.3 Last Updated By

Supply the name of the person who performed the most recent update to the use case description.

1.3.4 Date Last Updated

Enter the date on which the use case was most recently updated.

2. Use Case Definition

2.1. Actor

An actor is a person or other entity external to the software system being specified who interacts with the system and performs use cases to accomplish tasks. Different actors often correspond to different user classes, or roles, identified from the customer community that will use the product. Name the actor(s) that will be performing this use case.

2.2. Description

Provide a brief description of the reason for and outcome of this use case, or a high-level description of the sequence of actions and the outcome of executing the use case.

2.3. Preconditions

List any activities that must take place, or any conditions that must be true, before the use case can be started. Number each precondition. Examples:

1. User's identity has been authenticated.
2. User's computer has sufficient free memory available to launch task.

2.4. Postconditions

Describe the state of the system at the conclusion of the use case execution. Number each postcondition. Examples:

1. Document contains only valid SGML tags.
2. Price of item in database has been updated with new value.

2.5. Priority

Indicate the relative priority of implementing the functionality required to allow this use case to be executed. The priority scheme used must be the same as that used in the software requirements specification.

2.6. Frequency of Use

Estimate the number of times this use case will be performed by the actors per some appropriate unit of time.

2.7. Flow of Events

Provide a detailed description of the user actions and system responses that will take place during execution of the use case under normal, expected conditions. This dialog sequence will ultimately lead to accomplishing the goal stated in the use case name and description. This description may be written as an answer to the hypothetical question, "How do I <accomplish the task stated in the use case name>?" This is best done as a numbered list of actions performed by the actor, alternating with responses provided by the system.

2.8. Alternative Flows

Document other, legitimate usage scenarios that can take place within this use case separately in this section. State the alternative course, and describe any differences in the sequence of steps that take place. Number each alternative course using the Use Case ID as a prefix, followed by "AC" to indicate "Alternative Course". Example: X.Y.AC.1.

2.9. Exceptions

Describe any anticipated error conditions that could occur during execution of the use case, and define how the system is to respond to those conditions. Also, describe how the system is to respond if the use

case execution fails for some unanticipated reason. Number each exception using the Use Case ID as a prefix, followed by “EX” to indicate “Exception”. Example: X.Y.EX.1.

2.10. Includes

List any other use cases that are included (“called”) by this use case. Common functionality that appears in multiple use cases can be split out into a separate use case that is included by the ones that need that common functionality.

2.11. Special Requirements

Identify any additional requirements, such as nonfunctional requirements, for the use case that may need to be addressed during design or implementation. These may include performance requirements or other quality attributes.

2.12. Assumptions

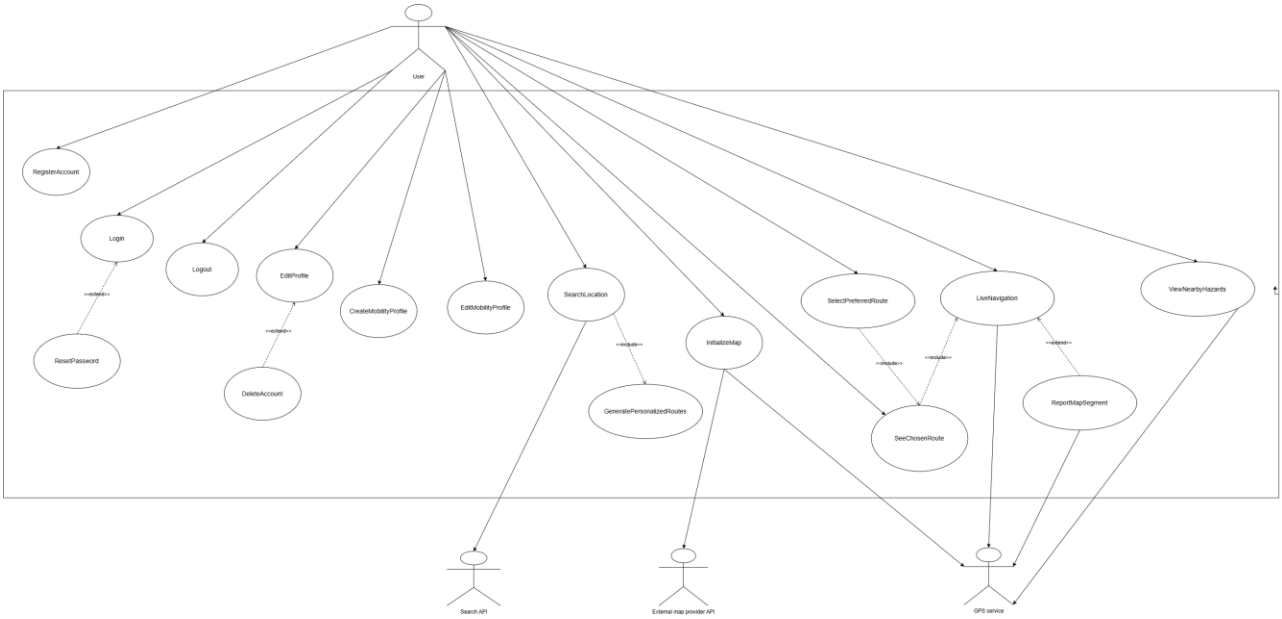
List any assumptions that were made in the analysis that led to accepting this use case into the product description and writing the use case description.

2.13. Notes and Issues

List any additional comments about this use case or any remaining open issues or TBDs (To Be Determineds) that must be resolved. Identify who will resolve each issue, the due date, and what the resolution ultimately is.

Use Case Model

Use Case Diagram



Use Case Descriptions

#1.1: Authentication

Use Case ID:	1.1-1		
Use Case Name:	RegisterAccount		
Created By:	Lin Ziqi	Last Updated By:	Lin Ziqi
Date Created:	03/09/2026	Date Last Updated:	03/09/2026

Actor:	User
Description:	Allows a new user to register an account using an email and password, verified via an email one-time code.
Preconditions:	The user has installed and launched the WalkingSG application and is on the registration interface.
Postconditions:	A new user account is successfully activated in the database, and the user is redirected to the login interface.
Priority:	High
Frequency of Use:	Low (once per user)
Flow of Events:	1. The user enters their email address, sets a password, and clicks "Register". 2. System validates that the password has greater or equal than 8 characters and includes at least one special character. 3. The system verifies that the provided email address is not already linked to an existing account in the database. 4. The system generates a 6-digit one-time code and sends it to the user's email address. 5. The user inputs the received 6-digit verification code into the verification screen and clicks "Verify". 6. The system validates the code, activates the account in the database, and navigates the user to the login screen.
Alternative Flows:	**AF-1 (Resend Verification Code):** At step 5, the user clicks "Resend Code". The system invalidates the previous code, generates a new 6-digit verification code, sends it to the user's email, and restarts the 10-minute expiry timer.
Exceptions:	**EX-1 (Weak Password):** At step 2, if the password is shorter than 8 characters or lacks a special character (@, #, $, %, ^, &, *), the system displays "Password must be at least 8 characters and contain at least one special character" and prompts the user to re-enter. **EX-2 (Duplicate Email):** At step 3, if the email already exists in

	the database, the system displays "Email already in use" and aborts registration. **EX-3 (Expired or Invalid OTP):** At step 6, if the 6-digit code is incorrect or exceeds the 10-minute expiration window, the system displays "Invalid or expired code" and prompts the user to re-enter or request a new code
Includes:	None
Special Requirements:	Passwords must be hashed using cryptography algorithm before database storage. The OTP must strictly expire 10 minutes after issuance (**REQ-AUTH-01.6**).
Assumptions:	The user has access to a functional email inbox to receive external mail.
Notes and Issues:	None

Use Case ID:	1.1-2		
Use Case Name:	Login		
Created By:	Lin Ziqi	Last Updated By:	Lin Ziqi
Date Created:	03/09/2026	Date Last Updated:	03/09/2026

Actor:	User
Description:	Authenticates a registered user using their email and password to grant access to the application.
Preconditions:	The user possesses an activated WalkingSG account and is not currently locked out.
Postconditions:	An active session token is issued, and the user is authenticated into the application.
Priority:	High
Frequency of Use:	High (daily/weekly)
Flow of Events:	1. The user enters their registered email address and password on the login screen. 2. The user selects the "Log In" button. 3. checks if the account is currently locked due to prior failed attempts. 4. The system validates the credentials against stored database records using cryptography verification. 5. The system establishes an active session token and navigates the user to the main map screen.
Alternative Flows:	**AF-1 (Forgot Password Path):** At step 1, the user selects "Forgot Password". The system navigates the user to the password recovery screen (triggers Use Case #1.1-3).
Exceptions:	**EX-1 (Invalid Credentials):** At step 4, if the email and password do not match database records, the system increments the failed login counter, denies access, and displays "Invalid email or password". **EX-2 (Account Lockout Triggered):** At step 4, upon the 5th consecutive failed login attempt, the system locks the account for 15 minutes and displays "Account locked for 15 minutes due to multiple failed attempts". **EX-3 (Account Already Locked):** At step 3, if the user attempts login while the 15-minute lockout timer is active, the system

	denies entry and displays "Account locked. Please try again later".
Includes:	None
Special Requirements:	Failed login attempts must be tracked in server-side cache with a 15-minute expiry time. Passwords must be masked on screen by default
Assumptions:	The user device has a stable internet connection.
Notes and Issues:	None

Use Case ID:	1.1-3		
Use Case Name:	ResetPassword		
Created By:	Lin Ziqi	Last Updated By:	Lin Ziqi
Date Created:	03/09/2026	Date Last Updated:	03/09/2026

Actor:	User
Description:	Allows a registered user who forgot their password to regain account access via a time-limited email reset link.
Preconditions:	The user has navigated to the "Forgot Password" screen from the login interface.
Postconditions:	The user's account password is updated in the database, and old credentials/tokens are invalidated.
Priority:	High
Frequency of Use:	Low
Flow of Events:	1. The user enters their registered email address and clicks "Send Reset Link". 2. System validates email existence, generates a secure reset token, and emails the link to the user 3. The user clicks the reset link in the email to open the password reset interface. 4. The user enters and confirms a new password that satisfies complexity rules (at least 8 characters and at least one special character). 5. The system validates the password, updates the database, invalidates the reset link, and redirects to the login screen.
Alternative Flows:	**AF-1 (User Aborts):** At step 1 or step 4, the user navigates back to the login screen. The system cancels the operation without updating credentials
Exceptions:	**EX-1 (Expired Reset Link):** At step 3, if the user opens the reset link more than 15 minutes after generation, the system displays "Password reset link has expired" and prompts the user to request a new link. **EX-2 (Password Fails Complexity):** At step 5, if the new password has fewer than 8 characters or lacks a special character, the system displays "Password does not meet requirements" and prompts for re-entry.
Includes:	None

Special Requirements:	Reset links must be single-use and invalidated immediately upon successful password change or after 15 minutes (**REQ-AUTH-03.2**).
Assumptions:	Mail delivery service operates within acceptable latency (< 60 seconds).
Notes and Issues:	None

Use Case ID:	1.1-4		
Use Case Name:	Logout		
Created By:	Lin Ziqi	Last Updated By:	Lin Ziqi
Date Created:	03/09/2026	Date Last Updated:	03/09/2026

Actor:	User
Description:	Terminates the user's active session and securely logs them out of the application
Preconditions:	The user is currently authenticated and logged in.
Postconditions:	The session token is permanently invalidated, and the application returns to the unauthenticated login screen.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. User navigates to the profile/settings interface and clicks "Log Out". 2. System prompts the user with a dialogue to confirm the logout action. 3. User confirms the logout action. 4. System invalidates the active session token, clears local user cache, and redirects the user to the login screen.
Alternative Flows:	**AF-1 (Cancel Logout):** At step 3, the user selects "Cancel". The system dismisses the prompt and maintains the active session on the current screen
Exceptions:	**EX-1 (Token Revocation Error):** At step 4, if the server fails to process session revocation due to connectivity drop, the system forcefully clears local tokens from device storage and redirects to the login screen.
Includes:	None
Special Requirements:	Must terminate session tokens immediately to prevent replay or unauthorised access.
Assumptions:	None
Notes and Issues:	None

Use Case ID:	1.1-5		
Use Case Name:	EditProfile		
Created By:	Lin Ziqi	Last Updated By:	Lin Ziqi
Date Created:	03/09/2026	Date Last Updated:	03/09/2026

Actor:	User
Description:	Allows a user to view and update their personal account information, specifically their display name.
Preconditions:	The registered user is logged in and viewing their user profile
Postconditions:	The user's updated personal information is saved to the database.
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. User selects "Edit Profile" on the profile interface. 2. System displays an editable form pre-populated with the user's current registered name and email. 3. User updates their name field and clicks "Save Changes". 4. System validates that the name input is non-empty and adheres to format constraints of length between 2 and 50 characters. 5. System commits the updated name to the database and displays "Profile updated successfully".
Alternative Flows:	**AF-1 (Discard Edits):** At step 3, the user clicks "Cancel" or taps the back icon. The system discards all modifications and restores the original profile view without updating the database.
Exceptions:	**EX-1 (Blank Name Field):** At step 4, if the user clears the name field and attempts to save, the system displays "Name cannot be left blank" and retains focus on the input field.
Includes:	None
Special Requirements:	In accordance with Business Rule 2.4, editing privileges must be restricted strictly to the user matching the unique User ID.
Assumptions:	Registered email address cannot be edited via the standard name edit form (managed separately under authentication).
Notes and Issues:	None

Use Case ID:	1.1-6		
Use Case Name:	DeleteAccount		
Created By:	Lin Ziqi	Last Updated By:	Lin Ziqi
Date Created:	03/09/2026	Date Last Updated:	03/09/2026

Actor:	User
Description:	Allows a user to permanently delete their account and remove all personal records from the system.
Preconditions:	The user is authenticated and navigating the account settings interface.
Postconditions:	The user account, credentials, and associated personal profile data are permanently erased from the database.
Priority:	Medium
Frequency of Use:	Very Low
Flow of Events:	1. The user selects "Delete Account" on the settings/profile screen. 2. The system displays a confirmation dialog warning that deletion is permanent and irreversible. 3. The user confirms the deletion request. 4. The system permanently deletes the user's account records, preferences, and personal information from the database. 5. The system terminates all active sessions and redirects the user to the welcome screen.
Alternative Flows:	**AF-1 (Cancel Deletion):** At step 3, the user selects "Cancel" on the confirmation prompt. The system dismisses the prompt and leaves the user account untouched
Exceptions:	**EX-1 (Database Deletion Failure):** At step 4, if a database timeout occurs, the system displays "Unable to delete account at this time. Please try again later" and retains all user data.
Includes:	None
Special Requirements:	Full data purge must comply with Singapore PDPA right-to-erasure and **REQ-AUTH-05.4**.
Assumptions:	None
Notes and Issues:	None

#1.2: Preference Management

Use Case ID:	1.2-1		
Use Case Name:	CreateMobilityProfile		
Created By:	Lin Ziqi	Last Updated By:	Lin Ziqi
Date Created:	03/09/2026	Date Last Updated:	03/09/2026

Actor:	User
Description:	Allows a newly registered user to set up their initial mobility constraints and environmental routing preferences for the first time.
Preconditions:	The user has registered and logged in and has not yet customised their default mobility profile.
Postconditions:	An initial personalised preference profile is established in the database under the user ID and bound to the routing engine.
Priority:	High
Frequency of Use:	Low (strictly once per user upon onboarding)
Flow of Events:	1. User navigates to the "Personal Declaration Form" in the profile section for the first time. 2. System initialises the form with default baseline values (mobility_needs as "None", Heat / UV sensitivity as "Medium"). 3. User configures their “age”, initial “mobility_needs” constraint, “heat_uv_sensitivity”, “max_stairs_per_route”, and “preferred_pace”. 4. User selects "Save Preferences" to confirm initial setup. 5. System commits the initial profile record to the database, activates routing engine binding, and displays "Profile created successfully".
Alternative Flows:	**AF-1 (Retain System Defaults):** At step 3, the user chooses to keep all default settings ("None" mobility_needs, "Medium" UV sensitivity) and selects "Save Preferences". The system records the default profile without modifications.
Exceptions:	**EX-1 (Setup Interruption):** At step 3, if the application loses connection before saving, the system preserves default parameters in the local session and prompts the user to submit again upon reconnection.
Includes:	None

Special Requirements:	Must comply with Singapore PDPA regulations regarding the collection and storage of user mobility and physical handicap data.
Assumptions:	Newly created user accounts are automatically initialised with database default values per **REQ-PREF-02.2** and **REQ-PREF-05.2**.
Notes and Issues:	None

Use Case ID:	1.2-2		
Use Case Name:	EditMobilityProfile		
Created By:	Lin Ziqi	Last Updated By:	Lin Ziqi
Date Created:	03/09/2026	Date Last Updated:	03/09/2026

Actor:	User
Description:	Allows an existing user to update their previously saved mobility constraints and routing preferences.
Preconditions:	The user has previously registered and configured an active mobility profile (Use Case #1.2-1 completed).
Postconditions:	Updated preferences are committed to the database, cached route data is flushed, and the routing engine immediately applies new parameters.
Priority:	Medium
Frequency of Use:	Low to Medium (occasional updates when physical condition or mobility constraint changes)
Flow of Events:	1. User opens the "Personal Declaration Form" in the profile tab. 2. System automatically pre-fills all form controls with the user's most recently saved database preferences. 3. User modifies their “mobility_needs”, “age”, “heat_uv_sensitivity”, “max_stairs_per_route”, or “preferred_pace”. 4. User selects "Save Preferences". 5. System commits the modified data to the database under the user ID. 6. System flushes locally cached route preferences, updates the routing engine parameters, and displays "Preferences updated successfully".
Alternative Flows:	**AF-1 (Discard Modifications):** At step 3, the user leaves the screen without saving. The system discards unconfirmed changes and keeps existing preferences intact.
Exceptions:	**EX-1 (Database Write Failure):** At step 5, if the database transaction fails, the system displays "Failed to update preferences. Please try again", retaining existing parameters.
Includes:	None

Special Requirements:	**REQ-PREF-05.2:** The system must immediately clear any locally cached route preferences and apply the updated variables to the routing engine for the very next query.
Assumptions:	Changes take effect immediately without requiring the user to log out and log in again.
Notes and Issues:	None

#1.3 Search & Map Interface

Use Case ID:	1.3-1		
Use Case Name:	InitializeMap		
Created By:	Benjamin	Last Updated By:	Benjamin
Date Created:	9/4/2026	Date Last Updated:	9/4/2026

Actor:	User
Description:	Renders the interactive map interface and handles initial GPS location permission prompts to auto-fill or clear the “Origin” input.
Preconditions:	Application is opened; Map API service is available.
Postconditions:	Interactive map is rendered, and initial “Origin” input field state is set.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System renders interactive map interface using external map provider API. 2. System allows user to zoom and pan on map interface. 3. System prompts user for device location (GPS) permissions upon entering main UI.
Alternative Flows:	**AF-1 (GPS Permission Granted):**System auto-fills "Origin" input field with user's current GPS location. **AF-2 (GPS Permission Denied):** System leaves "Origin" input field blank for manual entry.
Exceptions:	None
Includes:	None
Special Requirements:	Interactive map zooming and panning support (**REQ-MAP-01.1**).
Assumptions:	Device hardware supports GPS capabilities.
Notes and Issues:	None

Use Case ID:	1.3-2		
Use Case Name:	SearchLocation		
Created By:	Benjamin	Last Updated By:	Benjamin
Date Created:	9/4/2026	Date Last Updated:	9/4/2026

Actor:	User
Description:	Provides autocomplete suggestions for address searches and validates input coordinates against the Singapore bounding box before displaying markers.
Preconditions:	Map interface is loaded.
Postconditions:	Valid origin and destination coordinates are validated and displayed as markers on the map.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user inputs a search query into Origin/Destination fields. 2. The system provides autocomplete suggestions using external Search API. 3. The user selects origin and destination addresses. 4. System validates Origin and Destination coordinates against Singapore geographical bounding box. 5. The system displays markers for origin and destination on map interface. 6. The system calls **GeneratePersonalizedRoutes** to generate route variants based on the validated Origin and Destination.
Alternative Flows:	None
Exceptions:	**EX-1(Origin Coordinate Outside Bounding Box):** System rejects routing request and displays "Invalid location. Please select a valid origin within Singapore". **EX-2(Destination Coordinate Outside Bounding Box):** System rejects routing request and displays "Invalid location. Please select a valid destination within Singapore".
Includes:	**GeneratePersonalizedRoutes**
Special Requirements:	Hard geographical boundary checking restricted strictly to Singapore bounding box.
Assumptions:	External Search API is reachable.
Notes and Issues:	None

#1.4 Routing & Scoring Engine:

Use Case ID:	1.4-1		
Use Case Name:	GeneratePersonalizedRoutes		
Created By:	Huy	Last Updated By:	Huy
Date Created:	04/09/2026	Date Last Updated:	05/09/2026

Actor:	User (Internal Use Case of SearchLocation)
Description:	The system retrieves the user's routing preferences, processes validated coordinates to generate exactly 5 distinct route variants, calculates their metadata, and scores them.
Preconditions:	The user has submitted a valid Origin and Destination coordinate.
Postconditions:	A complete payload containing 5 scored route variants is returned to the UI.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The system receives the validated Origin and Destination coordinates. 2. The system retrieves the user's saved “heat_uv_sensitivity” and “mobility_needs” from the database, applying the defaults ("Medium" and "None") if no modifications exist. 3. The system generates exactly 5 distinct route variants: “Direct”, “Maximum Shelter”, “Barrier-Free”, “Low-Effort”, and “Wide/Quiet”. 4. The system calculates the total estimated walking time, and total physical distance for each route variant. 5. The system calculates a final match score for each route using the preferences retrieved in Step 2. 6. The system appends all calculated metadata and match scores to the route response payload and returns it to the client.
Alternative Flows:	**AF-1 (Hazard Filter):** At step 3, if the engine detects a map segment with a confidence score below -10, it filters that segment out of the "Barrier-Free Route" generation.
Exceptions:	**EX-2 (No Route):** At step 3, if no physical path exists between the origin and destination, the system returns a "No route available" error payload.
Includes:	None

Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

Use Case ID:	1.4.2		
Use Case Name:	SelectPreferredRoute		
Created By:	Fujita Natsuo	Last Updated By:	Fujita Natsuo
Date Created:	02/09/2026	Date Last Updated:	04/09/2026

Actor:	User
Description:	The user reviews the ranked route options and selects one for navigation.
Preconditions:	The 5 route variants carry final match scores
Postconditions:	A single route is locked in as the active navigation path.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The system displays the 5 route variants in the UI drawer, sorted by descending score 2. The system visually highlights the highest scoring route with a “Starred” badge 3. The user reviews the route and selects their preferred route.
Alternative Flows	**AF-1 (Tie-Breaker Sorting):** At step 1, if two or more routes share the same match score, the system prioritizes the route with the shorter estimated walking time. If still tied, it prioritizes the higher sheltered distance percentage. If both remain equal, the system defaults to the priority order: Maximum Shelter, Barrier-Free, Low-Effort, Wide/Quiet **AF-2 (Incomplete Metadata):** At step 1, if a route is missing distance or time data, the system displays "Distance information not available" or "Estimated travel time not available" on that specific card.
Exceptions	None
Includes	None
Special Requirements	None
Assumptions	The UI drawer can render all 5 route cards without requiring additional pagination.

Notes and Issues	None

Use Case ID:	1.4.3		
Use Case Name:	SeeChosenRoute		
Created By:	Huy	Last Updated By:	Huy
Date Created:	11/09/2026	Date Last Updated:	11/09/2026

Actor:	User
Description:	Displays the selected route as a full-screen polyline overlay on the map, allowing the user to visually inspect the path and direction before committing to live navigation.
Preconditions:	- The user has selected one route variant from the route results list (**Use Case #1.4-2** completed) - A valid route polyline payload exists for the selected route
Postconditions:	The selected route polyline is rendered on the map and the user either proceeds to live navigation or returns to the route list
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user taps on a route card from the route results list 2. The system renders the selected route as a polyline overlay on the full-screen map view, with origin and destination markers displayed 3. The system displays a summary panel showing the route's estimated time, total distance, sheltered percentage, stair count, and match score 4. The user taps the "Proceed" button to confirm the route selection 5. The system initiates live navigation for the confirmed route (**Use Case #1.5-1**)
Alternative Flows	**AF-1 (Return to Route List):** At step 4, the user taps the back icon. The system dismisses the route preview and returns the user to the route results list with all 5 route cards intact
Exceptions	**EX-1 (Route Payload Missing):** At step 2, if the route polyline data is unavailable or corrupted, the system displays "Route preview unavailable. Please try again" and

	returns the user to the route results list.
Includes	**LiveNavigation (#1.5-1)** – triggered upon user confirmation via the “Proceed” button
Special Requirements	The route polyline must be rendered within 1.0 second of the user tapping the route card
Assumptions	The route polyline coordinates are included in the route payload returned by **GeneratePersonalizedRoutes (#1.4-1)**
Notes and Issues	None

#1.5 Live Navigation

Use Case ID:	1.5-1		
Use Case Name:	LiveNavigation		
Created By:	Benjamin	Last Updated By:	Benjamin
Date Created:	04/09/2026	Date Last Updated:	04/09/2026

Actor:	User
Description:	Tracks user's real-time position during active navigation, provides dynamic directional text updates and trip metrics, automatically recalculates routes upon off-path deviation, and logs segment completion confidence scores upon destination arrival.
Preconditions:	Active route generated, GPS tracking enabled on device.
Postconditions:	Route marked as completed upon arrival within 5 meters, and confidence scores incremented.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System anchors map view to user's real-time GPS coordinate during active navigation. 2. The system displays text-based directional instruction and dynamically updates it based on user progression along route polyline. 3. The system continually monitors the user's deviation relative to the active route. 4. The system dynamically updates metric information (remaining distance and estimated time) as the user travels. 5. The system detects arrival when the user's GPS coordinate is within a 5-meter radius of the destination point. 6. System considers route completed and automatically increments confidence score of all traversed map segments by +1.
Alternative Flows:	None
Exceptions:	**EX-1(User Deviates > 20 Meters for 5 consecutive seconds):** System triggers automatic route recalculation using current GPS position as the new origin point.

Includes:	None
Special Requirements:	Dynamic update loop for distance/time metrics. Off-route threshold = 20 meters. Arrival radius = 5 meters. Increment value = +1.
Assumptions:	Device maintains active GPS tracking connection throughout navigation.
Notes and Issues:	None

#1.6 Community Hazard and Local Feed

Use Case ID:	1.6-1		
Use Case Name:	Report Map Segment		
Created By:	Hemanth	Last Updated By:	Hemanth
Date Created:	04/09/2026	Date Last Updated:	04/09/2026

Actor:	User
Description:	Allows a user to report a map segment as “Unwalkable” via the “Live Report” button.
Preconditions:	The system must have access to the user’s current GPS coordinate to identify the specific map segment that may be affected.
Postconditions:	The system must log the hazard directly to the “News/Hazards” screen. The affected map segment must have its confidence score decremented by 2 points.
Priority:	High
Frequency of Use:	Low - Medium (occurrence of hazards should not be too often)
Flow of Events:	1. The user clicks the “Live Report” button on his screen during active navigation. 2. The user reports his current location as “Unwalkable”, “Path blocked”, “Flooding”, “Something Else”, “Lift down” or “Broken Surface” on the app. 3. The system prompts the user for confirmation before submitting the report 4. The user confirms and submits the report. 5. The system maps the hazard to the specific route segment closest to user’s current GPS coordinate 6. The system deducts 2 points from that specific map segment's confidence score in the database, down to a clamp of -20. 7. The system adds the hazard to the “News/Hazards” screen to alert other users of the hazard.

Alternative Flows:	**AF-1 (Cancel Report):** Between step 2-4, the user selects "Cancel Report". The system dismisses the report and details filled so far and redirects the user back to his active navigation page.
Exceptions:	**EX-1 (Issue unselected):** If the user skips step 2, the system will prompt the user to select “Unwalkable”, “Path blocked”, “Flooding”, “Something Else”, “Lift down” or “Broken Surface” to confirm the severity of the hazard present and to declare the map segment has not viable.
Includes:	None
Special Requirements:	App support team may want to verify the presence of such hazards using public CCTV or site visits to avoid scams or acts of mischief by users.
Assumptions:	The user is honest and accurate in his description of the hazard and the hazard is truly severe enough to declare the map segment as inaccessible.
Notes and Issues:	None

Use Case ID:	1.6-2		
Use Case Name:	View Nearby Hazards		
Created By:	Hemanth	Last Updated By:	Hemanth
Date Created:	04/09/2026	Date Last Updated:	04/09/2026

Actor:	User
Description:	Displays a chronologically ordered list of currently reported hazards located within 5 kilometres of the user's position, so the user can be aware of unwalkable paths in their vicinity before setting out.
Preconditions:	User is authenticated and the device has supplied a current GPS location.
Postconditions:	A filtered hazard list displayed to the user. No system state is modified.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The user navigates to the "News/Hazards" screen. 2. The system queries the database for all map segments have received a confirmed hazard report within the past 72 hours, or with a negative confidence score. 3. The system filters the queried hazards to those within a 5-kilometre radius of the user's current GPS location. 4. The system sorts the filtered hazards chronologically, most recently reported first. 5. The system displays the sorted hazard list to the user.
Alternative Flows:	**AF-1 (No Hazards):** No hazards exist within the 5-kilometre radius. The system displays an empty-state message confirming no hazards are currently reported nearby.

Exceptions:	**EX-1 (GPS Location Unavailable):** If a user's current location has a weak signal or the user has denied permission to his GPS location, in both cases the current user location is unavailable, no radius can be computed. The system prompts the user to enable location access or enter a location manually.
Includes:	None
Special Requirements:	Feed must load within 3.0 seconds of the user opening the News screen. Radius is strictly 5 kilometres. Sort order is descending by report time in reverse chronological order with most recent reports appearing first.
Assumptions:	Every hazard report carries a timestamp, and every map segment carries a location that can be tested against the radius.
Notes and Issues:	None